Framework for Agent Gamification Based on Ontologies

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Contents

Introduction

Introduction

Assistant Professor at the University of Zagreb Faculty of Organization and Informatics, and a member of the Artificial Intelligence Laboratory at UNIZG FOI.

Scientific interests mainly in:

- multiagent systems,
- semantic modelling,
- gamification,
- artificial intelligence,
- computer games.

Introduction

One of the teachers of the following courses in Croatian or English:

- Multiagent Systems,
- Database Theory,
- Declarative Programming,
- Introduction to Artificial Intelligence,
- Introduction to Computer Games,
- Internet Security,
- Computer Game Development Platforms.

Introduction

Engaged in international activities and promoting international relations:

- Erasmus student at Karl-Franzens University of Graz (AT),
- Erasmus intern at Jožef Stefan Institute in Ljubljana (SI),
- Erasmus+ intern at Elettra Sincrotrone in Trieste (IT),
- 3-month research stay at Universitat Politècnica de València in Valencia (ES),
- ITEC student at Centre for Development of Advanced Computing in NOIDA (IN),
- 16-month research visit at Universitat Politècnica de València in Valencia (ES),

Introduction

Publications

Project activity

BSc and MA Mentees

Overview

Overview

	Part I	Part II
Phase A	MAS ontology	gamification ontology
Phase B	MAS framework	gamification framework

Table 1: Research in a nutshell: two parts, two phases each

- Phase 2 focuses on:
  - GIVEn - Games as Intelligent Virtual Environments Ontology
  - MAGO-Ga - Gamification of an Intelligent Virtual Environment Ontology
  - MAGO-Ga Framework - Implementation and case study

Methodology

- METHONTOLOGY ontology engineering methodology
- Structured approach with clearly defined steps:
  - Specification
  - Knowledge Acquisition
  - Conceptualisation
  - Formalisation
  - Integration
  - Implementation and Evaluation
- Applied to both GIVEn and MAGO-Ga ontologies

GIVEn Ontology

- **Purpose:** Provide concepts for describing video games as cro:IVEintelligent virtual environments (IVEs)
- **Intended use:** Modelling video game worlds using applicable concepts
- **Applications:**
  - Simulations
  - Enhancing reasoning capabilities of intelligent agents
  - Content generation (worlds, NPCs, items)
- **Level of granularity:** Abstract concepts that can be further specialized

GIVEn Knowledge Acquisition

- Analysis of existing video game ontologies:
  - Core Game Ontology (CGO)
  - Video Game Ontology (VGO)
- Concepts identified for modelling video games as IVEs:
  - NPCs, monsters, mobs
  - Places, points of interest
  - Game mechanics and dynamics

GIVEn Integration

- Integration with existing ontologies:
  - MAMbO5 - multiagent systems ontology
  - CGO - video game ontology
  - VGO - video game ontology
- Benefits:
  - Enhanced semantic reach
  - Improved reusability
  - Data consistency
  - Reduced redundancy

MAGO-Ga Ontology

- **Purpose:** Model gamified multi-agent systems as IVEs
- **Key components:**
  - Formal representation of gamification elements
  - Personality traits based on Five Factor model
  - Interactions between agents and gamification elements
- **Intended use:** Knowledge base for reasoning about gamified multi-agent systems

MAGO-Ga Framework

- GEAR - Gamified Emergent Agent Relations
- Implementation based on the recipeWorld model
- Key elements:
  - Service providers and consumers
  - Gamification techniques to influence agent behaviour
  - Personality profiles based on Five-Factor Model
  - Compatibility calculation between gamification techniques and agents
- Available on GitHub: `https://github.com/AILab-FOI/GEAR`

Case Study: The Recipe World

- Based on the recipeWorld model
- Agents as service providers and consumers
- Gamification techniques influence agent behaviour
- Personality profiles based on Five-Factor Model
- Compatibility between gamification techniques and agents

Implementation

- Both ontologies implemented using Protégé
- Available on GitHub repository
- OWL/XML serialization in appendices
- Consistent and provides additional knowledge when reasoned upon
- GEAR framework implements the concepts in a practical application

Evaluation

- Ontologies evaluated by domain experts
- Interview and review approaches
- Tested against specification documents
- Framework demonstrated through case study

Conclusion

- Phase 2 delivered:
  - GIVEn - Ontology for modelling video games as IVEs
  - MAGO-Ga - Ontology for gamified multi-agent systems
  - GEAR - Framework implementing the concepts
- Contributions:
  - Formal representation of gamification in multi-agent systems
  - Integration of personality traits with gamification techniques
  - Practical implementation demonstrating the approach

Bibliography

Bibliography i

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